

Solution

SOUND-04

Class 08 - Science

Section A

1.
(d) Increase
Explanation: The temperature increases, length of pendulum increases and hence time period of pendulum increases. Due to increment in its time period, a pendulum clock becomes slow in summer and will lose time.
2.
(a) Planting trees around roads
Explanation: To reduce noise pollution along roads, planting trees around roads is used. Plants are a good absorber of sound and reduce carbon dioxide concentration.
3.
(d) Communicate with one another
Explanation: Sound helps us to communicate with one another. We share our views and suggestions with others either by sound or through written medium.
4.
(a) 1320 m
Explanation: Speed of sound = 330 m/s, time = 4 seconds. Distance = speed $\times$ time = 330 m/s $\times$ 4 = 1320m. so, lightning occurs at a distance of 1320m.
5.
(a) Vocal cord
Explanation: The larynx, or voice box, is located in the neck and performs several important functions in the body. The larynx is involved in swallowing, breathing, and voice production. Sound is produced when the air which passes through the vocal cords causes them to vibrate and create sound waves in the pharynx, nose and mouth.
6.
(b) Frequency
Explanation: The number of oscillations per second of a vibrating object is called its frequency. Frequency is measured in Hertz (Hz).
7.
(c) quality
Explanation: quality
8.
(a) Defective speech
Explanation: Generally, a child with hearing impairment from birth has defective speech.
9.
(d) 12,000 times
Explanation: Frequency of sound = 200Hz, time = 1minute = 60 seconds. Number of vibrations = 60 $\times$ 200 = 12000times.
10.
(a) Higher than 20 kHz
Explanation: Ultrasound equipment is used for investigating and tracking medical problems. The sound of frequency greater than 20kHz is called ultrasound.
11.
(b) It is maximum in solids and minimum in gases
Explanation: It is maximum in solids and minimum in gases
12.
(a) 1.2 Hz
Explanation: 1.2 Hz
13.
(a) Sound
Explanation: Sound requires material medium for its propagation which may be solid, liquid or gaseous in nature. Light can travel in vacuum as no medium is required.
14.
(c) decibel (dB)

Explanation: The loudness of sound is measured in Decibel (dB). Sound of loudness greater than 80 dB is harmful for our ears.

15. (a) Sonometer

Explanation: Sonometer

16.

(b) Vibrate and produce sound

Explanation: When a tightly stretched band is plucked it vibrates and produces sound due to disturbance in medium.

17.

(d) Light travels faster than sound

Explanation: Lightning and thunder occur together but lightning is seen earlier than sound because light travels much faster than sound.

18. (a) 51 m

Explanation: 51 m

19.

(d) Drum

Explanation: Drum has a thin membrane of skin that vibrates on beating. The number of vibrations per second is less in the drum in comparison to whistle, flute, and sitar.

20.

(d) decrease its amplitude of vibration of the sound.

Explanation: Since loudness depends upon amplitude, so it can be increased by increasing amplitude and it can be decreased by decreasing amplitude.

21.

(b) Amplitude and loudness

Explanation: Loudness of sound depends upon amplitude of sound waves. Loudness of sound is proportional to amplitude of sound.

22.

(d) Sound in environment

Explanation: Frequency, reported in Hertz (Hz), measures the number of sound vibrations in one second. Amplitude, reported on the decibel (dB) scale, measures its pressure or forcefulness.

23.

(b) Oscillatory motion

Explanation: Vibration is also known as oscillatory motion as vibrating body oscillates to and fro. Oscillatory motion has fixed time-period.

24.

(b) i, ii & iii only

Explanation: Sound requires any medium to travel but in vacuum there is no medium, so sound cannot travel through them.

25. (a) Woman

Explanation: The voice of woman has higher frequency and shriller than man. We can easily identify the voice of female from male.

26.

(d) Time period

Explanation: The time taken by an object to complete one oscillation is called time-period. Time-period is reciprocal to frequency of sound.

27.

(d) Density of sea water is more

Explanation: Speed of sound increases with increase in density. So, the speed of sound in seawater is slightly more than distilled water because density of seawater is more due to presence of salts.

Section B

28.

(b) Both A and R are true but R is not the correct explanation of A.

Explanation: When an object vibrates, it causes movement in air molecules which further bump on to the other close air molecules as well. Due to these movements sound is produced.

29. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Shape of the outer part of ear is like funnel. When sound enters, it travels down a canal at the eardrum.

30. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Larger the amplitude, louder will be sound.

31.

- (d) A is false but R is true.

Explanation: Sound is produced by voice box or larynx which is present at the upper end of windpipe.

32.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: The S.I unit of frequency is hertz.

33.

- (d) A is false but R is true.

Explanation: Sound of trucks and buses are disturbing to ear so they are considered to be noise.

34.

- (d) A is false but R is true.

Explanation: Sound does need medium to travel. It does not travel in vacuum, as there is no medium.

35.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: These are the sound waves which can not be perceived by human ear and these waves are of very high energies.

36.

- (c) A is true but R is false.

Explanation: Pitch of sound depends upon frequency. Drum vibrates with low frequency and whistle has a high frequency. So drum has low pitched sound than whistle.

37.

- (c) A is true but R is false.

Explanation: Plantation on roadside will help to cut off the sound vibration produced by vehicles speaker by absorbing them and hence reduce noise pollution.

Section C

38. State True or False:

- (i) (b) False

Explanation: False; Speed of sound is less than speed of light in air.

- (ii) (a) True

Explanation: True

- (iii) (a) True

Explanation: True

- (iv) (a) True

Explanation: True

- (v) (b) False

Explanation: False

- (vi) (b) False

Explanation: False

- (vii) (a) True

Explanation: True

- (viii) (a) True

Explanation: True

- (ix) (b) False

Explanation: False; Sound cannot travel in vacuum

- (x) **(b)** False
Explanation: False
- (xi) **(a)** True
Explanation: True
- (xii) **(a)** True
Explanation: True. A medium is required for sound to travel.
- (xiii) **(a)** True
Explanation: True
- (xiv) **(b)** False
Explanation: False
- (xv) **(a)** True
Explanation: True
- (xvi) **(b)** False
Explanation: False
- (xvii) **(a)** True
Explanation: True
- (xviii) **(b)** False
Explanation: False
- (xix) **(a)** True
Explanation: True
- (xx) **(a)** True
Explanation: True
- (xxi) **(b)** False
Explanation: False
- (xxii) **(a)** True
Explanation: True
- (xxiii) **(a)** True
Explanation: True

Section D

39. Fill in the blanks:

- (i) 1. frequency
- (ii) 1. 10 dB
- (iii) 1. Amplitude
- (iv) 1. noise
- (v) 1. Noise
- (vi) 1. Time period
- (vii) 1. hertz
- (viii) 1. Hertz
- (ix) 1. Time period
- (x) 1. Frequency
- (xi) 1. Frequency
- (xii) 1. Amplitude
- (xiii) 1. 80 dB
- (xiv) 1. health
- (xv) 1. Vibrate